REP-086 · Full game audit and balance simulation

Game	Fruit Market 1.3.0 (rules version 6, protocol 3)
Report date	2026-11-05 (Asia/Kuala_Lumpur)
Type	Audit and recommendations. Nothing in the game's rules or balance was changed.
Request	"audit everything and suggest for improvements ... tabletop, UI and UX, game modes" and "a full simulation regarding balance ... suggest tweaks"
Status	Owner asked for the suggested changes (DEC-058, DEC-059). Built in 1.4.0 and 1.5.0: B1 to B4, B6, B7, U1 to U12, T1 to T4, T9, the rot stats, the print-kit items and a daily twist. Not built: B8 (no measurable gain), T3, T5, T8, and the Table Companion and other new modes.

1. The short version

1. **The core game is sound.** Seats are fair, the rules text matches the cards and numbers, and the basic game (no shopkeepers, favours or upgrades) is the tightest and fairest version.
2. **The extras are where the balance goes wrong.** Upgrades, shopkeeper perks and favour cards triple the number of lopsided games (11% in the basic game, 33% in the full game, 3 players).
3. **A rules mismatch:** the written rules, the set-up text and How to play all say "build one upgrade after each Market Day". The app lets a player build several in one evening. Skipping upgrades costs a player about $20 and nearly always loses (5.8% wins where 33% is fair).
4. **Two shopkeepers are off:** Big Basket (Benny) wins about 10 points above fair, Haggler (Rosa) about 9 points below. Extra starting cash does not help any perk, so cash perks do not work.
5. **Mango is the best fruit by 40%** (profit per fruit $5.30 against $3.7 for the others).
6. **Fruit almost never rots** when computers play (under 0.2 fruit per player per game). The headline tension, "sell it before it spoils", is not biting. Only human playtests can say whether this is true for people.
7. **Quick match is a 2-player game, the most lopsided size** (47% of games decided by $20 or more).
8. **The sales payout looked wrong on screen.** Totals jumped to their final value at the bell while customers were still being replayed. Fixed in a local branch (section 6).
9. **For a browser-first launch** the site has no install support (no manifest or service worker), no link-preview tags and no security headers. All three are small jobs.
10. **The printed kit cannot run the upgrade draft**: there are no upgrade cards, no day track and no queue strip, and the stall sheet cannot settle the tie-break (fruit sold).

2. How the simulation was run

- Every game is played start to finish by the computer players, using the same rules code as the app and server (all phases: buying, stocking, bell, rush, evening, night).
- **Baseline:** 3,000 games for each of 2, 3 and 4 players. Standard 10-day game, all players Hard, a different random shopkeeper for each stall (as at a table). Seats are fair, so any gap is the game, not the seat.

- **Other runs:** 500 to 1,500 games each. Game lengths, rule options, favour-card removal, upgrade policies, computer skill, and 11 changed-rule versions.
- **Reading the numbers:** win rates in 3-player games carry about +/-1.7 points of noise (500 games: +/-2.5). Treat gaps under 3 points as noise.
- **Limits:** these are computers, not people. They never play the Scout favour card, they buy only what they can sell, and their upgrade choices were tuned to the current app behaviour. Use the numbers to find structure, then confirm with human games.
- The test tooling was temporary and is not in the repo.

3. Baseline results (standard game)

	2 players	3 players	4 players
Win by seat	49.8 / 50.2	34.0 / 33.7 / 32.2	25.4 / 25.3 / 24.2 / 25.1
Median winning margin	$18	$13	$11
Margin as share of average final cash	12%	9%	7.6%
Middle half of margins	$9 to $32	$5 to $24	$5 to $21
Close games (margin $3 or less)	11%	17%	19%
Lopsided games (margin $20 or more)	47%	33%	28%
Average final cash	$150	$144	$144
Day-5 leader wins (fair: 50 / 33 / 25)	53.6%	33.7%	27.4%
Wins from last place at day 5	47%	34%	25%
Lead changes per game	3.4	4.8	5.6
Fruit sold per player	31.7	33.0	34.2
Demand served	76%	70%	69%
Fruit spoiled per player	0.14	0.08	0.05
Unsold ripe fruit at the end (jam maker pays $1)	3.2	2.1	1.6
Peak cash, one player / all together	$250 / $422	$258 / $639	$254 / $776

What this says:

- **Fair seats, no runaway leaders.** The leader on day 5 wins at chance in 3 and 4 players. The poorest-first buying order is a strong catch-up rule (see section 5, "Rotating order").
- **Smaller tables are swingier.** A 2-player day has one customer, so a few customers decide the game. Skill still shows (Hard beats Normal 79% in 2 players, 57% in 3 players against two Normals, and only 52% against three Easy in 4 players).
- **About 70% of demand is served in every setup.** Fruit supply, not cash, limits what stalls can do.
- **Paper money:** the printed bank (910)*coversthe*4 – *playerpeak*(776). It is tight if rich events stack up.

Shopkeepers (win rate, fair is 50 / 33.3 / 25)

Shopkeeper (perk)	2p	3p	4p	Gap to fair, 3p
Benny (Big Basket, +1 buy a day)	61.3	43.4	36.4	**+10.1**
Raj (Bargain Hunter)	52.1	36.2	27.7	+2.9
Kiran (Tip Jar)	49.4	34.9	24.3	+1.6
Saleh (Nest Egg, $15 start)	45.2	31.7	23.0	-1.6
Chen (Charmer, 3 favour cards)	48.5	29.9	22.6	-3.4
Rosa (Haggler, wins ties)	43.7	24.2	16.2	**-9.1**

The same order shows in all three table sizes, so it is the perks and not luck.

Fruit (profit per fruit bought, 3 players)

Blueberry $3.79, watermelon $3.73, apple $3.73, **mango $5.30**. Computers buy about 25 to 27 of each fruit per game, so everyone is fighting for mango. Mango customers pay up to $9 to $11 against a cost of $3 to $4.

Events (income per player on that day, against the average day, 3 players)

Festival +12.8, *HotelOrder* +7.2, Superfood +6.9, *MangoCraze* +5.4, Bad Harvest -0.3, *Heatwave* −1.4, Good Harvest -1.5, *PriceHike* −3.2, Minimum Price -3.3, *HealthInspector* −3.6, Blueberry Glut -3.8, *AppleBlight* −5.6, **Lorry Breakdown -$10.7**. Everyone feels an event, so events do not tilt a game. Without events total cash falls by 27%.

Customers (3 players)

- Richest per visit: Hotel Chef $46, Daniel $36, Gym Club $36, Rara Jenab $34, Ravi $34, John $33.
- Poorest per visit: Ben $19, Hafiz $20.5, Kumar $21, Aisyah $22, Hadi $22, Fatimah $22.
- Hardest to finish (share fully served): Ahmad 49%, Robert 51%, Wei Jie 57%, Yuki 57%, Hafiz 59%. All ask for 5 pieces.

Favour cards

- A player earns about 2 per game, and about 0.75 per player are never played.
- Win rate by cards earned: 0 to 1: 26%, 2 to 3: 35%, 4 to 5: 45% (3 players). Partly cause, partly good play.
- Removing all favour cards: lopsided games fall from 33% to 28% and average cash falls $2.4. No single card stands out.
- The computers never play Scout, so two of the 12 cards are dead in these numbers.

Game length and rule options (3 players unless marked, 500 to 600 games)

Option	Median margin	Lopsided	Close	Avg cash	Fruit spoiled
Standard (10 days)	$13	33.5%	16%	$146	0.08
8 days / 14 days	$13 / $15	31% / 38%	13% / 13%	$110 / $209	0.04 / 0.11

Option	Median margin	Lopsided	Close	Avg cash	Fruit spoiled
No events	$13	30%	16%	$107	0.03
No upgrades	$9	**16%**	21%	$146	**0.68**
No Market Days	$15	38%	16%	$121	0.13
Basic game (no perks, favours or upgrades)	**$7**	**11%**	**27%**	$141	0.69
Fixed prices	$14	36%	15%	$152	0.07
Price drift	$14	33%	14%	$160	0.08
No unripe fruit	$13	34%	15%	$149	0.08
2 players: standard / no upgrades / basic	$18 / $12 / $10	47% / 28% / 21%	12% / 19% / 18%	$152 / $153 / $148	0.18 / 1.17 / 1.23

Takeaways: upgrades add variance but no extra money on average (cash is the same with and without). The basic game is the fairest. Rot only appears once upgrades are gone.

Upgrades (3 players, one seat follows a fixed rule, 500 games)

Rule for that seat	Wins (fair 33.3)	Cash against the others
Computer's own choices	37.0%	+$1.5
Never builds any	**5.8%**	**-$20.2**
Builds everything it can	44.8%	+$6.2
Computer's choices, but never a Handcart	16.2%	-$12.6
... never Premium Display	27.6%	-$4.3
... never Cold Room	34.0%	+$0.4
... never Juice Bar	32.6%	-$0.7

- The Handcart is nearly mandatory. The Cold Room and Juice Bar change nothing, because computers almost never lose fruit to rot. Raising the Handcart price to $8 changed nothing (everyone can afford it by day 3).
- Players build 3.34 upgrades in 3 evenings. The rules say one per evening.

4. Rules and tabletop audit

Found now

1. **Upgrades per evening do not match the written rules.** RULES.md ("draft one stall upgrade"), the set-up screen ("build one") and How to play say one. The engine has no limit and the evening panel says "you may build upgrades". Fixing the app to one per evening is a one-line rule check, plus retuning the computer players. Measured effect: lopsided games 33% to 27%, never-build penalty -20*to*−10, fruit spoiled 0.08 to 0.33. Decision needed (section 8).
2. **The stall sheet has no "fruit sold" tally**, but the end-of-game tie-break is "whoever sold more fruit". Add a tally row.

3. **The printed kit is missing pieces the rules use:**
    - upgrade cards (4) and a draft procedure;
    - a day track with the Market Days and Closing Day marked;
    - the queue strip (dawn line-up) and the waiting pile area;
    - the fruit clock marker and day chips (the README says upgrade and Tip Jar tokens are "not printed yet");
    - a one-page crier card (the crier's top-minus-1 / top-minus-2 table) for the person who reads customers aloud.
4. **No solo or 2-player table rules** for a stall run by a card deck. The app has computer players; a table does not.
5. **No play-time estimate** for a table game. Time it at the first table test.
6. **Teaching ladder:** the simulation says the basic game is the fairest. Suggest teaching in four steps: basic, plus shopkeepers, plus favours, plus upgrades. The Playtest Pack could say so.
7. Stale code comment: `src/engine/game.ts` still says "rules version 4".

Checked and fine

Every count and number in RULES.md matches the app: 20 delivery, 36 customer, 13 event and 12 favour cards; start cash, shelf, buy limits, upgrade costs, jam price, Closing Day bonus, tip, generous customers list; Market Day days; tie-break order; walk-in counts.

5. Suggested balance changes (all need approval; effects are from the simulation)

#	Change	What the simulation showed	Notes
B1	**Upgrades: one per Market Day** (as written), and retune the computer to take the Handcart first	Lopsided 33% to 27%; median margin $13 to 11; *never* – *buildpenalty*−20 to -$10; rot 0.08 to 0.33. Average cash -5%	Matches every printed rule
B2	**Mango costs $1 more** (fresh $5, green $4)	Mango profit/fruit $5.30 to $4.57 (others $3.7); cash -4%; nothing else moves	$2 more would fully level it
B3	**Big Basket: the 5th piece costs $1 more**	Basket wins 43% to 39% (fair 33%); nothing else moves	Try $2 for fair
B4	**Rework Haggler, Nest Egg and Charmer so they give power, not cash**	Nest Egg at $10 start: 31.7% to 31.1% (no change). Haggler with $2 start: +2 points. Haggler's price counting $1 lower: no help	Cash is not scarce, so cash perks do nothing. Ideas, not yet simulated: Haggler "your first sale each day pays +$2"; Nest Egg "10-fruit shelf"
B5	**Keep the poorest-first buying order**	A rotating order doubles lopsided games (33% to 54%), halves close games (17% to 8%) and makes leaders win	The cost: an early lead means nothing (day-5 leader wins at chance)
B6	**2 players: add one walk-in**	Cash +45%; median margin unchanged in dollars, so 30% smaller as a share of cash	Does not cure lopsided games (47%)
B7	**Quick match and new players: offer the Basic**	Basic game: lopsided 11% (3p) / 21% (2p) against 33% / 47%	Easiest fairness win

#	Change	What the simulation showed	Notes
	game		
B8	**Cold Room and Juice Bar need a real job**	Neither changes win rate by more than +/-2 under any rule tested	Rot is rare, so insurance upgrades are worthless. Make them capacity or pricing upgrades
B9	Higher upgrade prices are not the answer	Prices x2.5: cash -16%, same upgrades built, Handcart still best	Do not use price as the lever

Not recommended: more starting cash, a bigger buy limit (3 to 5 changed nothing), and cheaper or dearer Handcart.

About rot: the rules say fruit goes bad, but computers lose under 0.2 fruit a game. Before changing any rule to make rot bite, record fruit spoiled and unsold per human game in the anonymous stats and look at a week of real play.

6. UI and UX audit

Checked on the live site at phone width (375 × 812) through home, set-up, morning, buying, stocking, pricing, the rush and the Wardrobe/Trophies/Settings screens.

Good

- Clear, warm art; the Showdown panels ("cheapest sells first") tell the story of each customer well.
- Market cells have full screen-reader labels ("Blueberry, fresh, $2, 6 left"); reduced motion is respected; undo purchases and Skip exist.
- The game screen has 327 page elements (it had 3,722 in October), so it is light.
- The page now has a "Skip tutorial" and a seven-card How to play.

To fix

#	Finding	Evidence	Suggested fix
U1	**Digits are misread in the pixel font at small sizes**	Rank "#2" looks like "#8"; the $2 blueberry price chip (9.9 px) looks like "$8"	Draw numbers under about 16 px in the rounded body font, and make price chips at least 12 px
U2	**Sales totals jumped to their final value at the bell**	Recorded cash: rivals' totals were final before the first customer; with the fix they climb customer by customer	**Done in branch `feat/payout-suspense` (local, not pushed)**; see below
U3	The rank chip has no spoken label and "#N" is unexplained	The accessibility tree shows no name for it	Label it "Rank 2 of 4"; show "2nd"
U4	Fast taps are dropped while buying	Three quick taps bought one blueberry	Queue taps, or shorten the lock and show a count on the cell
U5	Event preview wording	The morning card says "Tomorrow: Apple Blight. No apples arrive today"	Write event text that works for both days, or add a "tomorrow" version

#	Finding	Evidence	Suggested fix
U6	The set-up page is long and Start is at the bottom	Six shopkeepers, seats, length, four rules and three "Playtest options" in one scroll	Sticky Start button, three presets (Quick, Standard, With friends), and hide advanced options under "More rules"
U7	"Playtest options" is developer wording	Basic game, Fixed prices, Price drift	Rename "Beginner and expert rules"
U8	The "Psst..." hint on the pricing screen is partly under the Seal button	Screenshot	Add bottom padding equal to the button height
U9	Results have no cash-over-days chart and no share card (only the daily has one)	The data exists in the game history	Add a small chart and a share card
U10	The customer strip is clipped at the right edge with no hint to swipe	Screenshot	Add a fade or arrow
U11	No "change password" in Settings	AccountPanel has sign in, sign out, delete only	Add it, because there is no recovery
U12	Empty market slots draw ghost sprites (28 of them)	Screenshot	Show only slots that matter, or dim more

The payout change (U2)

- **What was wrong:** the game works out and pays every sale in one step at the bell (`resolve` in `src/engine/game.ts`). The screen drew that final cash straight away while the customers were replayed, so the numbers never built up.
- **What changed:** the cash chips and the rank badge are held back at their before-sales amount, and each customer's payment is added when its coins land, counting up a coin at a time. A Tip Jar tip lands at the end. Display only, no rule change.
- **Checked:** type check, lint, the whole test suite (174 tests, 3 new, including one that proves the held-back amounts equal what each stall had before the sales in 20 full games) and the build pass. In a real game the recorder showed the totals climbing in separate steps as each customer paid. Not yet tried online with a second player or on a phone.
- **Where:** commit `3394969` on branch `feat/payout-suspense` . Nothing is pushed.

7. Game modes, technology and operations

Modes today

Solo and pass-and-play (2 to 4 stalls), private online rooms, quick match (2 players), Daily market, Market League (6 stages, win to advance), Wardrobe and 43 achievements.

Mode ideas, best first

1. **Table Companion** (the app as the crier, bank and clock for a real table game). It uses the same rules code and fits the tabletop plan. It would also fix the table's biggest chore.

2. **League and Daily with objectives and stars** ("reach $120 with no upgrades", "sell 10 mangoes"), which adds goals without new rules.
3. **Daily with a twist of the day** (a rule option on rotation).
4. **Practice coach** (hints, "what would a good player do"), for new players.
5. **2 against 2** teams online; **async rooms** (a move a day); **replay links**.

Technology and operations

#	Finding	Fix
T1	No install support: no web manifest and no service worker	Add both (installable, full screen, works offline)
T2	No link-preview tags or page description	Add title, description and image tags so shared links look right
T3	Any unknown address (`/robots.txt` , `/sw.js` , `/manifest.webmanifest`) answers 200 with the game page	Add real files, and let unknown paths 404
T4	No security headers on the site	Add frame blocking, content-type and referrer headers, and a content security policy, through a `_headers` file
T5	Locking an account is per ID, so someone who knows an ID can lock it out for up to an hour	Lock by ID and address together, or add a delay for the same address
T6	Password hashing is 100,000 rounds, the Cloudflare maximum	Acceptable; no change possible
T7	Rate limit 12 account tries a minute per address	Fine; shared networks may hit it
T8	Dependencies: no known issues in shipped code; patch updates are available	Update in a quiet moment
T9	Tests and CI are strong (174 tests plus browser tests in CI)	Keep; add a test for the payout timing in the browser tests
T10	App code grew 40% since October (168 kB to 236 kB, 82 kB compressed)	Still small; watch it

8. Decisions needed from the owner

1. **Upgrades:** fix the app to one per Market Day (matches all printed text), or change the text to "as many as you can afford"? (B1)
2. **Which balance changes to try first:** B2 (mango), B3 (Basket) and B7 (Basic game for Quick match) are the safest.
3. **Shopkeeper rework (B4):** are you happy for me to propose new perks for Haggler, Nest Egg and Charmer?
4. **Rot:** add "fruit spoiled" and "unsold" to the anonymous stats so we can see how people really play?
5. **Table kit:** print upgrade cards, a day track, a queue strip and a crier card, and add a "fruit sold" row to the stall sheet?
6. **Push the payout change** (`feat/payout-suspense`) and the privacy-page change (`feat/privacy-page`), both still local.

7. **Public wording:** the About line in Settings still has your full name; keep or change?

9. Suggested order of work

1. Pixel-digit fix (U1), rank label (U3), tap queue (U4): small, no decisions.
2. Push the two finished branches and release 1.3.1 (privacy page and payout display).
3. Site basics (T1 to T4): install, link previews and headers.
4. Balance round 1 (B1, B2, B3, B7) after your approval, then re-run this simulation.
5. Table kit additions, then the first table test.
6. Table Companion mode.